

Self-Check Questions: Understanding Derivatives and Differentials

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Introduction

The purpose of these questions is to test and deepen an intuitive and rigorous understanding of derivatives, differentials, and related concepts in both single-variable and multivariable calculus. They are intended as self-check questions rather than exercises requiring lengthy computations.

1 Derivatives

1. What does it mean for a function to be differentiable at a point?

Answer. A function f defined near a point x is differentiable at x if there exists a linear map L_x (denoted df_x , or in one variable, multiplication by $f'(x)$) such that

$$f(x+h) = f(x) + L_x(h) + o(\|h\|) \quad \text{as } h \rightarrow 0,$$

i.e. the error in approximating $f(x+h)$ by $f(x) + L_x(h)$ vanishes faster than $\|h\|$ itself. In one variable this is equivalent to the existence of the limit $f'(x) = \lim_{h \rightarrow 0} (f(x+h) - f(x))/h$.

2. How is differentiability stronger than continuity?

Answer. If f is differentiable at x , then $f(x+h) - f(x) = df_x(h) + o(\|h\|) \rightarrow 0$ as $h \rightarrow 0$, since a linear map is bounded ($\|df_x(h)\| \leq \|df_x\| \|h\|$) and the remainder is $o(\|h\|)$; hence $f(x+h) \rightarrow f(x)$, i.e. f is continuous at x . The converse fails: $f(x) = |x|$ is continuous at 0 but has no derivative there, and $f(x) = x \sin(1/x)$ (with $f(0) = 0$) is continuous but not differentiable at 0, since its difference quotient does not converge.

3. How is differentiability stronger than the existence of partial derivatives?

Answer. If f is differentiable at x with differential df_x , then every partial derivative $\partial f / \partial x_i$ exists and equals $df_x(e_i)$, obtained by restricting attention to a single coordinate direction. But the mere existence of partial derivatives only says the restrictions of f to each coordinate axis are differentiable; it says nothing about behavior along other directions. Example: $f(x, y) = xy/(x^2 + y^2)$ for $(x, y) \neq (0, 0)$, $f(0, 0) = 0$. Both partials at the origin exist (and equal 0), yet f is not even continuous there (approaching along $y = x$ gives $1/2$), so f cannot be differentiable at the origin.

4. Why is differentiability fundamentally about approximation rather than slopes?

Answer. The defining property is the existence of a linear map df_x making the error $f(x+h) - f(x) - df_x(h)$ vanish faster than h ; “slope” is merely the coefficient that happens to characterize the unique linear map when there is one variable. In higher dimensions there is no single number that could be called “the slope” – only the approximating linear map has invariant meaning – so approximation, not slope, is the primitive notion, and it is the one that generalizes.

5. What is the geometric meaning of the derivative?

Answer. $f'(x)$ is the slope of the unique line through $(x, f(x))$ that best approximates the graph of f near x , namely the tangent line $y = f(x) + f'(x)(x' - x)$. Equivalently, it measures the instantaneous rate of change of f at x .

6. What is the relationship between differentiability and tangent lines?

Answer. The tangent line at x is exactly the graph of the affine function $x' \mapsto f(x) + f'(x)(x' - x)$ – the best affine approximation to the graph of f , in the sense that the vertical distance between graph and line is $o(|x' - x|)$. A tangent line fails to exist when f is not differentiable at x : at a corner (like $|x|$ at 0, where the left and right slopes disagree), or at a point of vertical tangency (like $x^{1/3}$ at 0, where the difference quotient diverges rather than converging to a finite slope).

7. What changes when we move from one variable to several variables?

Answer. The definition of differentiability keeps the same form – a linear map L_x with $f(x+h) - f(x) - L_x(h) = o(\|h\|)$ – but L_x can no longer be described by a single number, since h now ranges over $\mathbb{R}^n$. Concretely, the difference quotient $(f(x+h) - f(x))/h$ no longer makes sense, because division by a vector is undefined; the definition must instead be phrased in terms of an approximating linear map (or, in coordinates, a full Jacobian matrix) rather than a single limiting ratio.

8. Why is the derivative of a multivariable function no longer a number?

Answer. A single number times a vector h always points in the direction of h itself, so it cannot capture rates of change that differ from direction to direction. The differential must, in general, respond differently to different displacements h at the same point, which requires a genuinely linear map – encoded, once bases are chosen, by a matrix – rather than a scalar multiplier.

9. What object replaces the slope in several variables?

Answer. The linear map df_x replaces the slope; once coordinates are fixed it is represented by the Jacobian matrix, and for a scalar-valued function it is often re-expressed via the gradient vector $\nabla f(x)$, with $df_x(h) = \nabla f(x) \cdot h$ once an inner product is chosen.

10. Why is the derivative considered the best linear approximation?

Answer. df_x is the unique linear map L_x for which

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x) - L_x(h)}{\|h\|} = 0,$$

i.e. among all linear maps, df_x is the one whose error term vanishes strictly faster than $\|h\|$; any other linear map leaves an error that stays comparable to $\|h\|$ rather than becoming negligible relative to it.

2 The Differential

11. What is a differential?

Answer. The differential of f at x is the linear map df_x that best approximates the increment of f near x , characterized by $f(x+h) = f(x) + df_x(h) + o(\|h\|)$. It exists precisely when f is differentiable at x .

12. Why is the differential a function rather than a number?

Answer. df_x takes a vector h as input and produces the corresponding linear approximation to the increment, $df_x(h)$; it is thus a map, not a single numeric value. Only after specifying both the point x and a displacement h does one obtain an actual number (or vector), $df_x(h)$.

13. Why do we write $df_x(h)$ instead of simply $df(h)$ or $df(x, h)$?

Answer. The linear map applied to h depends on the base point x — it is generally a different map at different points — so the subscript records which map is in use. Writing $df(h)$ alone would hide this dependence and wrongly suggest a single, universal linear map; $df_x(h)$ makes clear that df_x is the map and h is its argument, rather than treating x and h as two arguments on equal footing.

14. What information is contained in the subscript x ?

Answer. It records the base point at which the increment of f is being approximated, selecting which of the (generally point-dependent) linear maps df_x is meant.

15. What information is contained in the argument h ?

Answer. h is the displacement — direction and magnitude — away from the base point x ; $df_x(h)$ approximates $f(x+h) - f(x)$ for that particular displacement.

16. Why are both x and h required to specify a value of the differential?

Answer. x fixes which linear map is being used, and h fixes which input is fed into that map. Changing x can change the map itself (a different point can have a different derivative), while changing h only changes the output of the same fixed map; both pieces of data are needed to determine the number $df_x(h)$.

17. What is the domain of the linear map df_x ?

Answer. The space of displacements h from x : for $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ this is $\mathbb{R}^n$, or more intrinsically the tangent space $T_x \mathbb{R}^n \cong \mathbb{R}^n$.

18. What is the codomain of the linear map df_x ?

Answer. The target space of f , namely $\mathbb{R}^m$, or more intrinsically the tangent space $T_{f(x)} \mathbb{R}^m$.

19. What exactly is meant by calling df_x a linear approximation to f near x ?

Answer. It means the affine function $h \mapsto f(x) + df_x(h)$ matches $f(x+h)$ up to an error that is $o(\|h\|)$. The word “linear” refers to the fact that the correction term $df_x(h)$ is linear in h , even though the full approximating expression $f(x) + df_x(h)$ is, strictly speaking, affine (linear plus the constant $f(x)$).

20. Can df_x differ from $df_{x'}$ at two different points x and x' ?

Answer. Yes, in general df_x and $df_{x'}$ are different linear maps for $x \neq x'$; e.g. for $f(x) = x^2$, $df_x(h) = 2xh$, so the multiplier $2x$ changes with x . The one exception is when f is itself affine, in which case df_x is the same linear map at every point.

21. If f is itself a linear map, how does df_x relate to f ?

Answer. If f is already affine, $f(x) = Ax + b$, then $df_x = A$ for every x : the differential coincides with f ’s linear part at every base point, independent of x , because an affine map is already its own best affine approximation everywhere – the error term is identically 0, not merely $o(\|h\|)$.

22. What is the differential of a constant function?

Answer. If $f(x) = c$ for all x , then $df_x = 0$ (the zero linear map) at every x , since $f(x+h) - f(x) = 0$ exactly.

23. What is the differential of the identity function?

Answer. If $f(x) = x$, then $df_x(h) = h$ for every x ; the differential is the identity map at every point.

24. What is the differential of $f(x) = x^2$?

Answer. $f(x+h) = x^2 + 2xh + h^2 = f(x) + 2xh + h^2$, and since $h^2 = o(h)$, the linear part is $df_x(h) = 2xh$. So df_x is multiplication by $2x$, consistent with $f'(x) = 2x$.

3 The Meaning of dx

25. What is dx ?

Answer. dx denotes the differential of the identity function $x \mapsto x$; as computed above, dx is the linear map $h \mapsto h$, i.e. $dx(h) = h$ at every point. It is conventionally used as the name of the “input variable” h when writing other differentials, as in $dy = f'(x) dx$.

26. Is dx a number?

Answer. No. dx is a linear map (the identity map on displacements); once an input h is supplied, $dx(h) = h$ is a number, but the symbol dx by itself denotes the map, not a value.

27. Is dx an infinitesimal?

Answer. No, not in the standard framework used throughout this document: there is no “infinitely small nonzero number” involved. dx is simply the name given to the ordinary, finite

displacement h fed into a linear map. The historical infinitesimal language can be made rigorous only within nonstandard analysis, a separate framework not used here.

28. Is dx a variable?

Answer. Yes, in the sense that $dx = h$ stands for an arbitrary displacement ranging over all real numbers (or vectors); it plays the role of the free argument of the linear map df_x .

29. Does the symbol dx denote the same mathematical object regardless of the base point x at which the derivative is evaluated?

Answer. Yes. Since dx is the differential of the identity function, $dx(h) = h$ at every base point – exactly the special case, described above, of an affine (here linear) function being its own differential everywhere. So “ dx ” unambiguously means “the identity map on displacements,” independent of which point we differentiate at.

30. Why can dx be chosen arbitrarily?

Answer. $dx(h) = h$ simply returns whatever displacement h is supplied, with no constraint tying it to a particular value; any real number (or vector) may be substituted. This is what allows $dy = f'(x) dx$ to be understood as an identity between linear maps, holding for every choice of input.

31. What role does dx play in $dy = f'(x) dx$?

Answer. dx plays the role of the arbitrary input h ; the equation is shorthand for the equality of linear maps $dy(h) = f'(x) \cdot dx(h) = f'(x)h$, i.e. it states how dy acts on an arbitrary displacement, expressed via the identity map dx .

32. Can dx be negative?

Answer. Yes; $dx(h) = h$ can take any real value, positive, negative, or zero, since h may be a displacement in either direction.

33. Can dx equal zero?

Answer. Yes, $dx = 0$ corresponds to $h = 0$, the trivial displacement, for which every differential necessarily returns $df_x(0) = 0$ by linearity.

34. Can dx be large?

Answer. Yes; as an algebraic object there is no restriction on the size of h , so $dx(h) = h$ for any h , arbitrarily large included.

35. Why is dx usually assumed to be small?

Answer. The practical use of $dy = f'(x) dx$ is to approximate the actual increment $\Delta y = f(x + h) - f(x)$; this approximation is accurate only when h (hence dx) is small, because the error is $o(h)$ – meaningfully negligible only relative to a small h . Smallness is a condition for dy to be a good numerical estimate of Δy , not a requirement for the identity $dy = f'(x) dx$ itself.

36. Does the mathematical definition require smallness?

Answer. No. The equation $dy = f'(x)dx$, understood as an identity between linear maps, holds exactly for every h , large or small. Smallness matters only when dy is being used to approximate Δy numerically.

37. In rigorous calculus, what replaces the vague idea of an infinitesimal?

Answer. The $o(\|h\|)$ error term in the definition of differentiability: it rigorously captures “negligible compared to h ” via an ordinary ε – δ limit,

$$\lim_{h \rightarrow 0} \frac{\|f(x+h) - f(x) - df_x(h)\|}{\|h\|} = 0,$$

without invoking any nonstandard numbers.

4 The Meaning of dy

38. What is dy ?

Answer. dy is the differential of f at x , i.e. $dy = df_x$, viewed as a linear map of the displacement dx ; concretely $dy(dx) = f'(x)dx$ in one variable, or $df_x(h)$ in general.

39. How is dy computed?

Answer. In one variable, $dy = f'(x)dx$: multiply the derivative at x by the chosen displacement dx . In several variables, $dy = df_x(h) = \sum_i \partial f / \partial x_i(x) dx_i$: contract the gradient (or Jacobian) with the displacement vector.

40. Why does dy depend on both x and dx ?

Answer. $dy = f'(x)dx$ depends on x because the derivative $f'(x)$ generally varies from point to point, and on dx because that is precisely the input to the linear map; changing either the base point or the displacement changes the resulting value of dy .

41. Is dy an approximation?

Answer. Yes; dy is designed to approximate the true change in the function’s value resulting from moving from x to $x + dx$.

42. What quantity does dy approximate?

Answer. $\Delta y = f(x + dx) - f(x)$, the actual change in the function’s output.

43. What is the relationship between dy and Δy ?

Answer. $\Delta y = dy + o(dx)$ as $dx \rightarrow 0$; that is, dy is the linear (leading-order) part of Δy , and the two differ by an error vanishing faster than dx itself.

44. When are dy and Δy close?

Answer. Whenever dx is small and f is differentiable at x ; the smaller $|dx|$, the smaller the relative error between dy and Δy , since that error is $o(dx)$.

45. Why are they generally not equal?

Answer. $\Delta y = f(x+dx) - f(x)$ reflects the full nonlinear behavior of f , whereas $dy = f'(x) dx$ captures only the linear (first-order) term. Unless f is itself affine, these differ by a nonzero nonlinear remainder whenever $dx \neq 0$.

46. Can dy be computed without evaluating the original function?

Answer. Yes; $dy = f'(x) dx$ requires only the derivative $f'(x)$ at the single point x and the chosen dx – no evaluation of f at the shifted point $x+dx$ is needed, unlike Δy , which requires computing $f(x+dx)$ directly.

47. Why is dy easier to compute than the exact change?

Answer. Computing $\Delta y = f(x+dx) - f(x)$ may require evaluating a complicated function at a new point, whereas dy only needs the already-known derivative $f'(x)$ multiplied by dx – a single multiplication rather than a full function evaluation and subtraction.

5 Linear Maps

48. Why must df_x , the best local approximation to f near x , be linear rather than merely affine?

Answer. An affine approximation would be $A(h)+c$ for a constant c . Since $f(x+h)-f(x) \rightarrow 0$ as $h \rightarrow 0$, the approximation must match this limit, which forces $c = 0$. Among affine maps that approximate the increment $f(x+h) - f(x)$ to the required order, only the linear ones ($c = 0$) survive, which is why df_x is required to be linear rather than merely affine.

49. What two algebraic properties – additivity and homogeneity – together define a linear map?

Answer. A map L is linear if it satisfies additivity, $L(h_1 + h_2) = L(h_1) + L(h_2)$, and homogeneity, $L(\lambda h) = \lambda L(h)$ for scalars λ ; together these are equivalent to $L(\lambda_1 h_1 + \lambda_2 h_2) = \lambda_1 L(h_1) + \lambda_2 L(h_2)$.

50. How is linearity used when approximating nonlinear functions?

Answer. Even though f itself may be highly nonlinear, near any fixed point x its behavior is, to leading order, governed by the linear map df_x . This lets complicated nonlinear reasoning near x – estimating sensitivities, solving perturbed equations – be replaced by ordinary linear algebra (matrix operations, superposition), valid for small displacements from x .

51. In what precise sense can a differentiable function be approximated locally by its differential, and how small is the resulting error term?

Answer. For h near 0, $f(x+h) = f(x) + df_x(h) + r(h)$ where $r(h) = o(\|h\|)$, meaning $\|r(h)\|/\|h\| \rightarrow 0$ as $h \rightarrow 0$; equivalently, for every $\varepsilon > 0$ there is $\delta > 0$ such that $\|r(h)\| \leq \varepsilon\|h\|$ whenever $\|h\| < \delta$. So the error is not merely small but small *relative to* $\|h\|$ – it shrinks faster than the displacement itself.

52. What is the relationship between differentials and matrices?

Answer. Once bases are chosen for the domain and codomain, any linear map between finite-dimensional spaces is represented by a matrix; since df_x is linear, it is represented, in the standard bases, by the Jacobian matrix of f at x .

53. When is df_x represented by a 1×1 matrix, i.e. a single number?

Answer. When $f : \mathbb{R} \rightarrow \mathbb{R}$ is a scalar function of a single scalar variable; then df_x is multiplication by the single number $f'(x)$.

54. When is df_x represented by a $1 \times n$ row vector?

Answer. When $f : \mathbb{R}^n \rightarrow \mathbb{R}$ is scalar-valued but depends on several variables; then $df_x(h) = \nabla f(x) \cdot h$ is represented by the row vector of partial derivatives $(f_{x_1}, \dots, f_{x_n})$.

55. When is df_x represented by a general $m \times n$ matrix?

Answer. When $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ with both n and m possibly greater than 1 (a vector-valued function of several variables); then df_x is represented by the full $m \times n$ Jacobian matrix, with (i, j) entry $\partial f_i / \partial x_j(x)$.

56. How does the Jacobian matrix represent df_x in the standard bases?

Answer. The j -th column of the matrix is $df_x(e_j) = \partial f / \partial x_j(x)$; assembling all n columns gives the Jacobian matrix $J_f(x)$, and $df_x(h) = J_f(x)h$ for every h .

57. Why is the Jacobian only a representation of the differential rather than the differential itself?

Answer. The differential df_x is an intrinsic linear map between vector spaces (or tangent spaces), independent of any choice of coordinates. The Jacobian matrix is what results from choosing bases in the domain and codomain to write that map as an array of numbers; a different choice of bases produces a different matrix representing the very same map df_x . So the Jacobian is basis-dependent data attached to df_x , not df_x itself.

6 Single-Variable Calculus

58. Why is the equation $dy = f'(x) dx$ simply a restatement, in differential notation, of $df_x(h) = f'(x)h$?

Answer. By definition $dy = df_x$, and dx is the differential of the identity function, i.e. the name given to the input h . Substituting these names into $df_x(h) = f'(x)h$ gives exactly $dy = f'(x) dx$; the two expressions describe the same linear map, merely written with different notation for the input (h versus dx).

59. Why does this look like ordinary multiplication?

Answer. Because both dy and dx , evaluated at a given h , are ordinary real numbers ($dx(h) = h$, $dy(h) = f'(x)h$), and $dy = f'(x) \cdot dx$, read pointwise, states that one number equals $f'(x)$ times another – indistinguishable in appearance from ordinary real-number multiplication.

60. Is it actually multiplication?

Answer. At the level of values, yes: for fixed h , $dy(h)$ really is the product of the number $f'(x)$ and the number $dx(h) = h$. But $dy = f'(x) dx$ is best understood as an identity between two linear maps – dy , and the map dx scaled by the constant $f'(x)$ – which is scalar multiplication of a linear map, a related but distinct notion from multiplying two independent numbers.

61. Why does the notation nevertheless work so well?

Answer. Scalar multiplication of the linear map dx by the number $f'(x)$ coincides, value by value, with ordinary multiplication of real numbers, so all the familiar algebraic rules for products carry over unchanged. This is why manipulating dy and dx as if they were ordinary numbers under multiplication almost never leads to an error, even though the underlying objects are maps.

62. When is manipulating differentials algebraically justified?

Answer. Whenever the manipulation can be translated back into a true statement about the corresponding linear maps or limits of difference quotients: e.g. treating dy/dx as a ratio of numbers is fine because it always equals the well-defined derivative $f'(x)$; a product-rule computation $d(uv) = u dv + v du$ is fine because it is exactly the linearization of the derivative product rule.

63. Why does $\frac{dy}{dx} = f'(x)$ behave like an ordinary fraction?

Answer. For fixed nonzero h , $dy(h)/dx(h) = f'(x)h/h = f'(x)$, a genuine ratio of two real numbers equal to the derivative regardless of which nonzero h is chosen. Since this ratio is literally computed by dividing one number by another, the usual rules of fraction algebra apply.

64. In what situations would treating it as an ordinary fraction be incorrect?

Answer. In several variables, where dy is really $df_x(h) = \nabla f(x) \cdot h$ and there is no single “ dx ” to divide by (h is a vector, and division by a vector is undefined). Attempting to treat partial-derivative notation as a literal fraction of infinitesimals, or canceling differentials across unrelated equations, can lead to incorrect results in that setting.

65. What is the rigorous meaning of $\frac{dy}{dx}$?

Answer. Notation for the derivative $f'(x)$ itself, defined as the limit $\lim_{h \rightarrow 0} (f(x+h) - f(x))/h$; equivalently, the unique scalar such that $dy(h) = f'(x) \cdot dx(h)$ for every h — a limit/linear-map statement, not literally a quotient of two independently meaningful infinitesimal quantities.

7 Multivariable Calculus

66. What is the differential of $f(x, y)$?

Answer. $df_{(x,y)}(dx, dy) = f_x(x, y) dx + f_y(x, y) dy$, where f_x, f_y are the partial derivatives: the linear map sending a displacement (dx, dy) to the corresponding linear approximation of the change in f .

67. Why is there a single differential df_x at each point, rather than a separate differential for each independent variable x and y ?

Answer. The differential is, by definition, a single linear map on the full displacement vector $h = (dx, dy)$ — a joint object capturing how f responds to simultaneous changes in both variables, including cross-effects between them, not a pair of independent one-variable differentials. Splitting it into separate per-variable differentials would only recover the two partial derivatives, losing information about f 's behavior along directions that mix both variables.

68. What is the role of dx and dy in $df = f_x dx + f_y dy$?

Answer. dx and dy are the components of the displacement vector $h = (dx, dy)$ – the differentials of the coordinate functions x and y – serving as the two independent inputs whose linear combination, weighted by f_x, f_y , produces the value of df at that displacement.

69. Why are dx and dy independent?

Answer. dx and dy are the coordinate projections of an arbitrary displacement $h = (h_1, h_2) \in \mathbb{R}^2$ ($dx(h) = h_1$, $dy(h) = h_2$); since h ranges freely over all of $\mathbb{R}^2$, h_1 and h_2 may be chosen independently, with no constraint linking them.

70. What is the meaning of the vector (dx, dy) ?

Answer. It represents an arbitrary displacement h in the plane – the amounts by which the first and second coordinates change – and serves as the general input to the differential $df_{(x,y)}$.

71. Why does df accept a vector instead of separate numbers?

Answer. A change of position in the plane is inherently two-dimensional – moving away from (x, y) is specified by a full displacement vector, not by two unrelated numbers – and the differential must be evaluated at a single such displacement to produce the corresponding first-order change in f .

72. How does the Jacobian act on (dx, dy) ?

Answer. For $f : \mathbb{R}^2 \rightarrow \mathbb{R}^m$, the Jacobian $J_f(x, y)$ is an $m \times 2$ matrix, and

$$df_{(x,y)}(dx, dy) = J_f(x, y) \begin{pmatrix} dx \\ dy \end{pmatrix},$$

ordinary matrix-vector multiplication of the Jacobian with the displacement vector.

73. What is the geometric meaning of df ?

Answer. $df_{(x,y)}$ describes the tangent plane to the graph of f at $(x, y, f(x, y))$: the plane $z = f(x, y) + f_x(x, y) dx + f_y(x, y) dy$ is the affine approximation whose graph best matches the surface $z = f(x, y)$ near that point.

74. How is the gradient related to the differential?

Answer. For scalar-valued f , once an inner product (e.g. the standard dot product) is chosen on the domain, there is a unique vector $\nabla f(x)$ such that $df_x(h) = \nabla f(x) \cdot h$ for all h ; the gradient is that representing vector.

75. Why is the gradient not the same object as the differential?

Answer. df_x is a linear functional – an object that eats a vector and returns a number – living in the dual space, whereas $\nabla f(x)$ is a vector living in the same space as the displacements h . They are related by the inner product, $df_x(h) = \nabla f(x) \cdot h$, but are not literally the same kind of mathematical object.

76. Why does the gradient depend on the chosen inner product while the differential does not?

Answer. df_x is defined purely from the limiting behavior of f and requires no additional structure. The gradient, however, is defined as the vector representing df_x via an inner product; changing the inner product changes which vector v satisfies $\langle v, h \rangle = df_x(h)$ for all h , so $\nabla f(x)$ depends on that choice even though df_x does not.

8 The Chain Rule

77. What happens to differentials under composition?

Answer. If $k = f \circ g$ at a point x , then $dk_x = df_{g(x)} \circ dg_x$: the differential of a composition is the composition of the individual differentials, evaluated at the appropriate points.

78. Why is the chain rule simply composition of linear maps?

Answer. Since g is well approximated near x by dg_x and f is well approximated near $g(x)$ by $df_{g(x)}$, composing these two best linear approximations gives the best linear approximation to $f \circ g$ near x ; this composition $df_{g(x)} \circ dg_x$ is precisely what the chain rule computes.

79. How does the Jacobian matrix encode the chain rule?

Answer. Composition of linear maps corresponds, once matrix representations are chosen, to matrix multiplication, so the Jacobian of $f \circ g$ at x is the matrix product $J_f(g(x)) J_g(x)$.

80. Why do Jacobian matrices multiply?

Answer. Because the differential of a composition is the composition of the differentials — a fact intrinsic to linear maps — and the matrix representing a composition of linear maps is exactly the product of the matrices representing each map individually.

81. What is the differential of $f(g(x))$?

Answer. $d(f \circ g)_x(h) = df_{g(x)}(dg_x(h))$: first apply dg_x to h , then feed the result into $df_{g(x)}$.

82. How does this explain repeated applications of the chain rule?

Answer. For a composition of several functions $f_1 \circ f_2 \circ \cdots \circ f_k$, iterating the two-function rule shows the overall differential is the composition of all the individual differentials in order, and correspondingly the overall Jacobian is the product of all the individual Jacobian matrices in the same order — matching the familiar rule of “multiplying the derivatives along the chain” from single-variable calculus.

9 Implicit Functions

83. Why can we differentiate both sides of $F(x, y) = 0$?

Answer. If $y = y(x)$ is a differentiable function satisfying $F(x, y(x)) = 0$ identically on some interval, the left- and right-hand sides define the same (constant, zero) function of x , so their differentials at any point must agree. Differentiating both sides is simply applying the chain rule to this identity in x .

84. What exactly is being differentiated?

Answer. The composite function $x \mapsto F(x, y(x))$, obtained by substituting the assumed implicit solution $y(x)$ into F – not $F(x, y)$ as a function of two independent variables.

85. What assumptions on F are required for it to be differentiable at the point in question, so that dF_x exists?

Answer. F must be differentiable (e.g. continuously differentiable, $F \in C^1$) in a neighborhood of the point (x_0, y_0) , so that its differential $dF_{(x,y)} = F_x dx + F_y dy$ exists there.

86. How does the differential lead to implicit differentiation?

Answer. Treating x and $y(x)$ together as a curve through (x, y) -space, the total differential of F along this curve is $dF = F_x dx + F_y dy$. Since F is identically 0 along the curve, this differential must vanish, giving $F_x dx + F_y dy = 0$, which leads directly to the implicit-differentiation formula.

87. Why does

$$F_x dx + F_y dy = 0$$

follow?

Answer. F is constant (equal to 0) along the implicitly defined curve, so its differential along any displacement (dx, dy) tangent to that curve must be 0. But $dF = F_x dx + F_y dy$ by the general formula for the differential of a two-variable function, so $F_x dx + F_y dy = 0$.

88. Under what condition on F_y may we solve this equation for dy/dx ?

Answer. We need $F_y \neq 0$ at the point in question; only then can $F_x dx + F_y dy = 0$ be solved uniquely for dy in terms of dx (namely $dy = -\frac{F_x}{F_y} dx$), and hence for the ratio dy/dx .

89. How does this produce

$$\frac{dy}{dx} = -\frac{F_x}{F_y}?$$

Answer. Starting from $F_x dx + F_y dy = 0$ and assuming $F_y \neq 0$, isolate dy : $dy = -\frac{F_x}{F_y} dx$; dividing by dx gives $\frac{dy}{dx} = -\frac{F_x}{F_y}$.

90. How is this connected to the Implicit Function Theorem?

Answer. The Implicit Function Theorem states that if F is C^1 near (x_0, y_0) , $F(x_0, y_0) = 0$, and $F_y(x_0, y_0) \neq 0$, then there exists a differentiable function $y = y(x)$ near x_0 with $F(x, y(x)) = 0$, and $y'(x) = -F_x/F_y$. The condition $F_y \neq 0$ identified above – needed to solve for dy/dx – is precisely the hypothesis of this theorem guaranteeing that $y(x)$ exists and is differentiable in the first place.

10 Common Misconceptions

91. Is dx the same thing as Δx ?

Answer. For the independent variable, they are conventionally taken to be equal ($dx := \Delta x$); this is a convention, not a theorem, that makes $dy = f'(x) dx$ directly comparable to $\Delta y = f(x + \Delta x) - f(x)$. For the dependent variable, however, dy and Δy are *not* equal in general – dy is only the linear approximation to Δy .

92. Is df_x defined only when f is differentiable at x , or can the symbol be written down even when it is not?

Answer. df_x is defined precisely when f is differentiable at x ; if not, no linear map satisfies the required approximation property, and the symbol has no meaning. One could still formally assemble a “candidate” linear map from partial derivatives, but as the $xy/(x^2 + y^2)$ example shows, such a candidate can fail to actually approximate f to the required order.

93. Is $df_x(h)$ a number, or is df_x itself – before being applied to h – a number?

Answer. $df_x(h)$ is a number (or vector, for vector-valued f): the output of feeding a specific displacement into the linear map. df_x itself, without an argument, is not a number but the linear map as a whole.

94. In the single-variable case, is $f'(x)$ itself a linear map, or is it the number that determines the linear map df_x ?

Answer. $f'(x)$ is a number. The linear map is df_x , defined by $df_x(h) = f'(x)h$; $f'(x)$ is the scalar by which df_x multiplies its input, not the map itself.

95. Does the differential df_x depend on a choice of basis or coordinates, even though its Jacobian matrix representation does?

Answer. No. df_x is an intrinsic linear map defined independently of any coordinate system. Its matrix representation, the Jacobian, depends on the chosen bases for domain and codomain – a change of basis transforms the matrix – but the underlying map df_x is unchanged by such a choice.

96. Are differentials defined only for scalar-valued functions?

Answer. No; the definition $f(x+h) = f(x) + df_x(h) + o(\|h\|)$ makes sense for any $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, with df_x a linear map into $\mathbb{R}^m$. Scalar-valued functions ($m = 1$) are just the special case where df_x can additionally be represented by a gradient vector via an inner product.

97. Why do physics and engineering texts often speak about infinitesimals?

Answer. The infinitesimal picture (dx as an “infinitely small” change) is an intuitive, computationally convenient heuristic inherited from the historical development of calculus; it mirrors physical reasoning about small perturbations and often leads quickly to correct relationships, such as by “canceling” differentials, without explicitly invoking limits or $o(\cdot)$ notation.

98. Why does the infinitesimal intuition often produce correct formulas?

Answer. The heuristic manipulations – treating dx, dy as numbers, canceling them, chaining derivatives – can, in essentially every standard case, be rigorously justified by translating them into statements about the actual linear maps df_x and their $o(\|h\|)$ error terms. The heuristic tracks genuine linear-algebraic structure even though it does not name it explicitly, which is why it so reliably gives correct final answers.

99. Which statements about differentials are rigorous mathematics, and which are heuristic shortcuts?

Answer. Rigorous: the existence of df_x with error $o(\|h\|)$; the chain rule as composition of linear maps; the identity $dy = f'(x) dx$ as an equality of linear maps for every h ; the Implicit Function Theorem. Heuristic: describing dx as an “infinitely small” number; “canceling” differentials as if they were literal small physical quantities without reference to the underlying limit; treating dy and Δy as interchangeable for non-small dx . The heuristic language is a convenient shorthand for, not a substitute for, the rigorous definitions.

11 Connections to Later Mathematics

100. How do differentials generalize to functions between arbitrary vector spaces?

Answer. For $f : X \rightarrow Y$ between normed vector spaces, f is (Fréchet) differentiable at x if there is a bounded linear map $L_x : X \rightarrow Y$ with $\|f(x+h) - f(x) - L_x(h)\|_Y = o(\|h\|_X)$ as $h \rightarrow 0$ – exactly the same defining property as the finite-dimensional differential, phrased in general normed spaces.

101. What is the Fréchet derivative?

Answer. The bounded linear map Df_x (if it exists) satisfying $f(x+h) - f(x) - Df_x(h) = o(\|h\|)$; it is the direct generalization of the differential to general normed spaces.

102. How is the elementary differential a special case of the Fréchet derivative?

Answer. Taking $X = \mathbb{R}^n$ and $Y = \mathbb{R}^m$ in the Fréchet-derivative definition recovers df_x exactly: a linear map on finite-dimensional space is automatically bounded (continuous), and the defining approximation property is identical, so the elementary differential is simply the Fréchet derivative specialized to Euclidean spaces.

103. How are differentials related to tangent vectors?

Answer. A curve $\gamma(t)$ through x with $\gamma(0) = x$ has velocity vector $\gamma'(0)$, a tangent vector at x ; the differential maps tangent vectors at x to tangent vectors at $f(x)$ via $df_x(\gamma'(0)) = (f \circ \gamma)'(0)$, i.e. df_x is precisely the map induced on tangent spaces by f (the pushforward).

104. What are cotangent vectors?

Answer. Elements of the dual space of the tangent space at a point – linear functionals that eat tangent vectors and return numbers. The space of all such functionals at x is the cotangent space $T_x^* \mathbb{R}^n$.

105. Why is the differential naturally a covector?

Answer. For scalar-valued f , $df_x : T_x\mathbb{R}^n \rightarrow \mathbb{R}$ is a linear functional on the tangent space at x — it eats a tangent vector h and returns a number — which is exactly the definition of a cotangent vector at x . So df_x lives naturally in the cotangent space, not the tangent space.

106. How does the gradient arise from the differential via the Riesz representation theorem (the “musical isomorphism”)?

Answer. The Riesz representation theorem says that on a finite-dimensional inner product space, every linear functional φ can be uniquely written as $\varphi(h) = \langle v, h \rangle$ for some vector v in the space itself. Applying this to the covector df_x gives a unique vector $\nabla f(x)$ with $df_x(h) = \langle \nabla f(x), h \rangle$ for all h . This vector-from-covector correspondence is the musical isomorphism (raising an index, $\sharp$) between cotangent and tangent spaces induced by the inner product, and it is exactly how the gradient is produced from the differential.

107. How do differential forms generalize differentials?

Answer. A differential 1-form on a manifold is a smooth assignment, to each point x , of a covector $\omega_x \in T_x^*M$, not necessarily arising as df_x for a single global function f . Differentials df of functions are the simplest (“exact”) examples of 1-forms; higher-degree differential forms further generalize this by allowing alternating multilinear (rather than merely linear) functionals on several tangent vectors at once.

108. Why is df considered a differential 1-form?

Answer. At each point x , df_x is a linear functional on the tangent space $T_x\mathbb{R}^n$ (a covector), and the assignment $x \mapsto df_x$ varies smoothly with x whenever f is smooth. A smooth field of covectors is by definition a differential 1-form, so df qualifies — indeed the prototypical example, called an exact 1-form.

109. How does this viewpoint unify calculus, linear algebra, and differential geometry?

Answer. Calculus supplies the analytic notion of approximation (the $o(\|h\|)$ limit) defining the differential in the first place; linear algebra supplies the language — linear maps, dual spaces, matrices, inner products — used to describe and compute with df_x once it exists; and differential geometry extends the pointwise linear-algebraic picture (tangent/cotangent spaces, covectors, 1-forms) coherently over an entire manifold. This shows that the elementary “best linear approximation” idea from single-variable calculus is really the local, linearized shadow of a global geometric structure.